RETAILPULSE 360

Sales, Profit & Customer Intelligence System

Project Report

MainCrafts SkillSprint — Data Science & Data Analytics Using Python

Candidate: [Your Name] | Date: [Submission Date]

Executive Summary

Over 2021–2024, the business processed 9,994 orders from 793 customers, generating $6.26M in sales and $533.4K in profit — an overall profit margin of 8.5%. Sales were essentially flat year over year, so profit growth must come from mix and discount discipline rather than assumed top-line growth.

Technology leads both sales ($2.97M) and profit ($391.0K). Furniture sells almost as much as Technology ($2.28M) but returns only a fraction of the profit ($44.5K). Discounting is strongly linked to eroding profit (correlation -0.344); average profit turns negative once discount reaches roughly 30%. 29 products are unprofitable overall, concentrated in Tables and Chairs. Regional performance is balanced. Only 3 of the top 10 sales customers are also top 10 by profit.

Management should investigate why Furniture consistently loses money, why top-revenue customers are frequently not top-profit customers, and why deep discounts persist on categories that cannot absorb them — and should consider the five recommendations detailed at the end of this report.

1. Project Title & Problem Statement

Project Title: RetailPulse 360 — Sales, Profit & Customer Intelligence System

A growing retail company has thousands of transactions but does not have a clear understanding of what is actually driving its business performance. Management needs answers on which products, regions, segments and customers drive revenue and profit, whether discounting is helping or hurting profitability, how sales move over time, and what concrete actions the data supports.

2. Objective

- Understand and clean a real-world retail transactions dataset.
- Engineer analytical fields (year, month, shipping duration, profit margin).
- Calculate core business KPIs (sales, profit, orders, customers, AOV, margin).
- Analyze sales, profitability, customers, products, regions, time trends, discounting and shipping.
- Translate findings into business-ready insights and evidence-based recommendations.

3. Dataset

The analysis uses a Global-Superstore-schema retail transactions dataset covering 2021–2024, with 10,017 rows and 19 columns spanning Order ID, Order/Ship Date, Ship Mode, Customer, Segment, Region, Category, Sub-Category, Product, Sales, Quantity, Discount, Profit and Order Priority.

Column	Meaning	Type
Order ID / Date / Ship Date	Order identity and timing	Text / Date
Ship Mode / Order Priority	Fulfillment method and urgency	Categorical
Customer ID / Name / Segment	Who purchased, and customer type	Text / Categorical
Region / State / Country	Where the order shipped	Categorical
Category / Sub-Category / Product	What was purchased	Categorical / Text
Sales / Quantity / Discount / Profit	Transaction financials	Numeric

4. Methodology

The project followed a structured analytics workflow: Understand → Clean → Analyze → Visualize → Discover → Explain → Recommend, executed in Python (Pandas, NumPy, Matplotlib, Seaborn) inside a single analysis notebook.

- Data loading and inspection (shape, types, missing values, duplicates)
- Data cleaning (date conversion, de-duplication, missing-value handling)
- Feature engineering (Year, Month, Order_Month, Shipping_Days, Profit_Margin)
- KPI calculation (Sales, Profit, Orders, Customers, AOV, Margin)
- Exploratory analysis across Sales, Profitability, Customers, Products, Regions, Time, Discounting and Shipping
- Business interpretation of every analytical result, followed by insight synthesis and recommendations

5. Data Cleaning

Missing values found: Ship Mode: 20, Order Priority: 45

Duplicate rows found and removed: 23

Rows before cleaning: 10,017

Rows after cleaning: 9,994

Order Date and Ship Date were converted to proper datetime values. Fully duplicated rows were removed since they carried no additional information. Missing Ship Mode and Order Priority values were filled with the most common / neutral category rather than dropped, preserving the sales and profit figures those rows carried. No rows or columns were removed without a documented reason.

6. Analysis & Visualizations

6.1 Key Business KPIs

KPI	Value
Total Sales	$6.26M
Total Profit	$533.4K
Total Orders	9,994
Total Customers	793
Units Sold	50,311
Average Order Value	$627
Overall Profit Margin	8.5%

6.2 Sales Analysis

Technology is the leading category by sales ($2.97M); Office Supplies is the lowest ($1.01M).

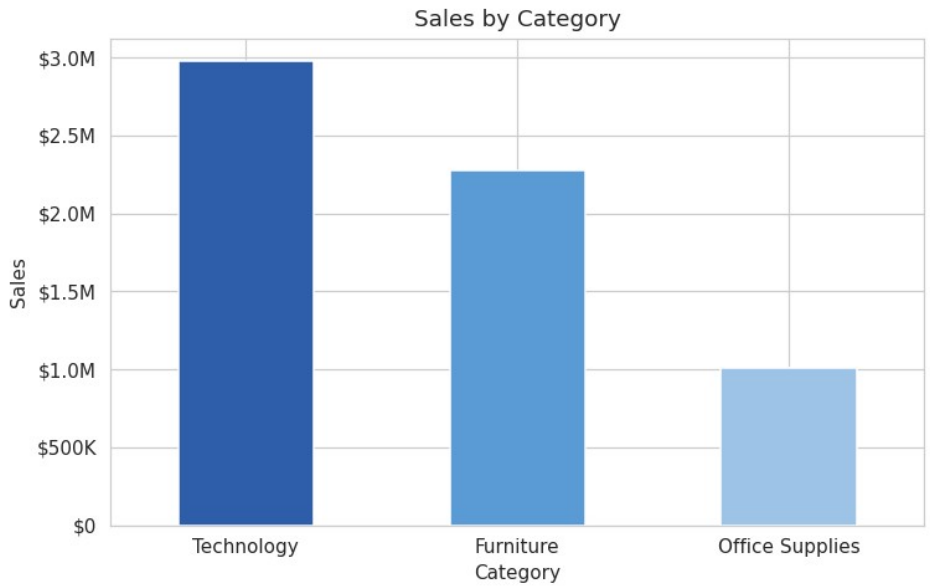


Figure 1. Sales by Category

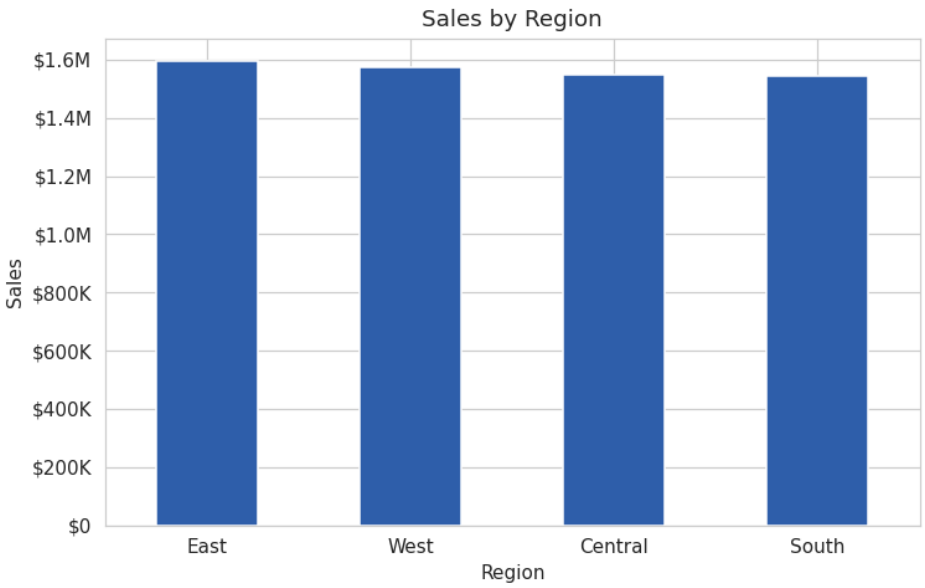


Figure 2. Sales by Region

6.3 Profitability Analysis

Technology converts sales to profit far more efficiently (13.1% margin) than Furniture (2.0%) or Office Supplies (9.7%). The sales-leading region and profit-leading region are not the same, confirming profit should be tracked separately from revenue.

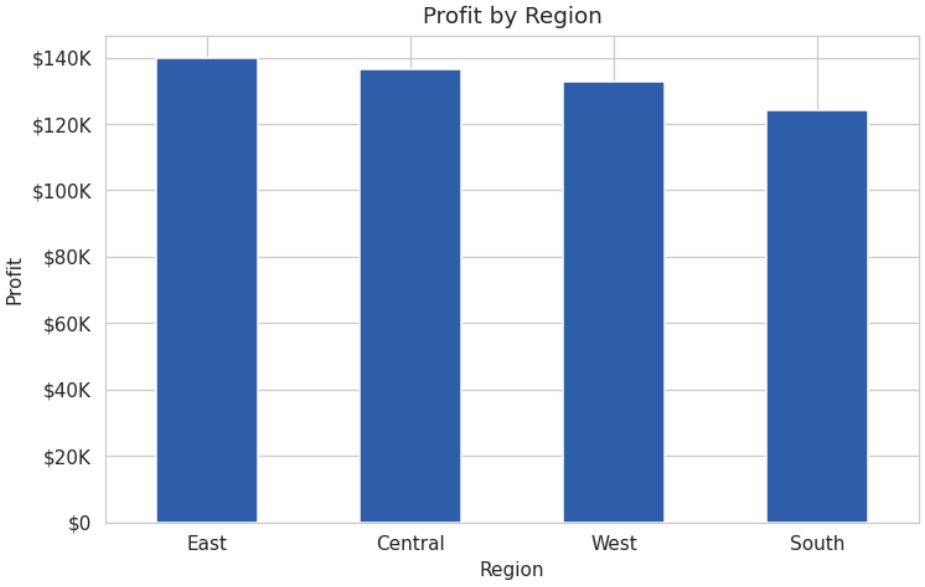


Figure 3. Profit by Region

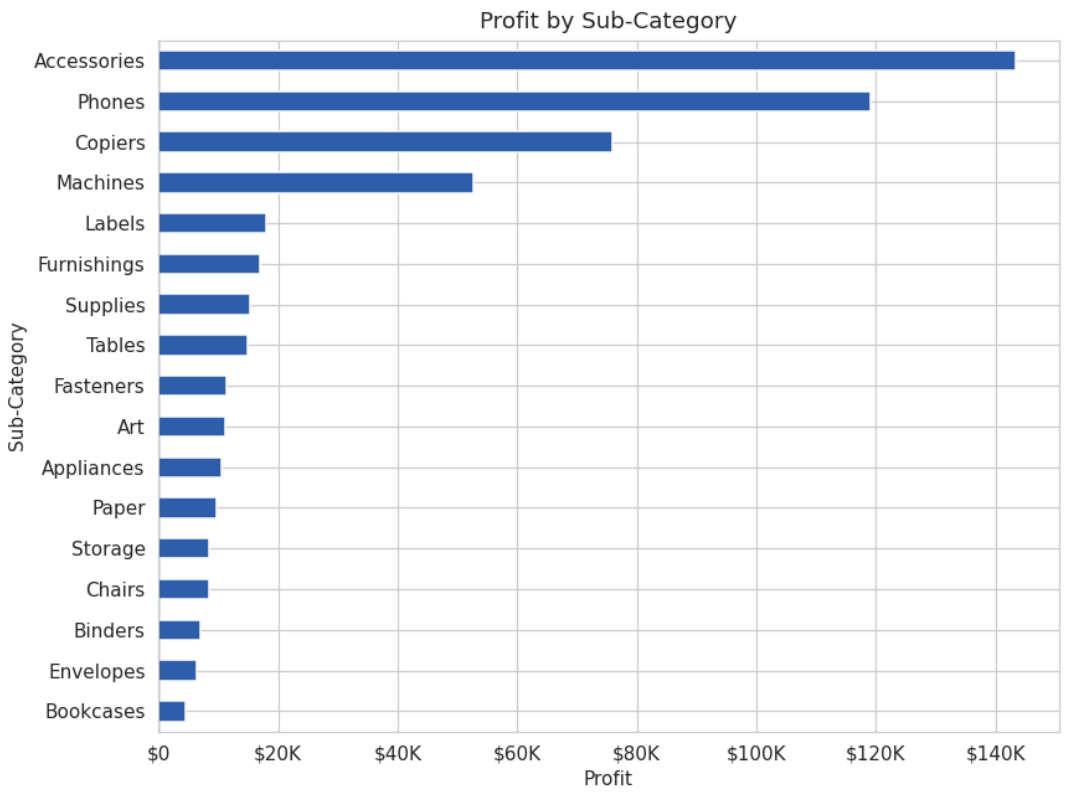


Figure 4. Profit by Sub-Category

Note: every sub-category is net profitable in aggregate — the losses below are concentrated in specific products within Tables and Chairs, not the sub-categories as a whole.

6.4 Loss-Making Products

29 products are unprofitable overall. The ten biggest loss-makers (combined -$7.7K) are all Tables and Chairs items carrying above-average discounts — the losses trace to discount depth on specific SKUs, not to weak demand.

Product	Category	Sales	Profit	Avg. Discount
Tables Item 1073	Furniture	$20.8K	-$1.2K	34%
Chairs Item 1031	Furniture	$49.9K	-$1.1K	23%
Tables Item 1082	Furniture	$24.3K	-$1.1K	24%
Furnishings Item 1126	Furniture	$18.2K	-$850	25%
Bookcases Item 1011	Furniture	$37.0K	-$731	24%
Tables Item 1077	Furniture	$24.6K	-$668	22%
Tables Item 1081	Furniture	$21.4K	-$662	17%
Tables Item 1091	Furniture	$21.3K	-$611	23%
Tables Item 1071	Furniture	$24.8K	-$421	25%
Tables Item 1068	Furniture	$23.7K	-$418	21%

6.5 Discount vs. Profit

The correlation between discount and profit is -0.344 — moderately negative. Average profit per order is positive up to roughly 20% discount and turns negative from about 30% onward. This is an association, not proof of causation, since discount depth also correlates with product category.

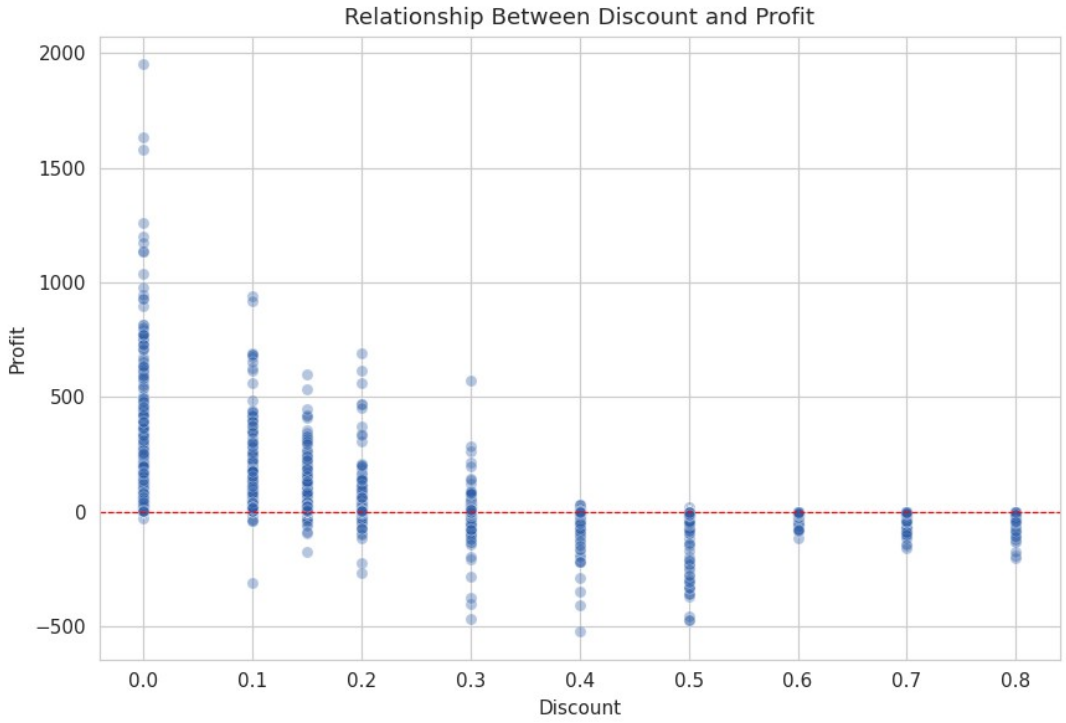


Figure 5. Discount vs. Profit (scatter, sampled orders)

6.6 Time-Series Analysis

Sales moved only -4.8% from 2021 ($1.62M) to 2024 ($1.54M) — essentially flat. March is the strongest calendar month ($576.7K); February the weakest ($464.8K).

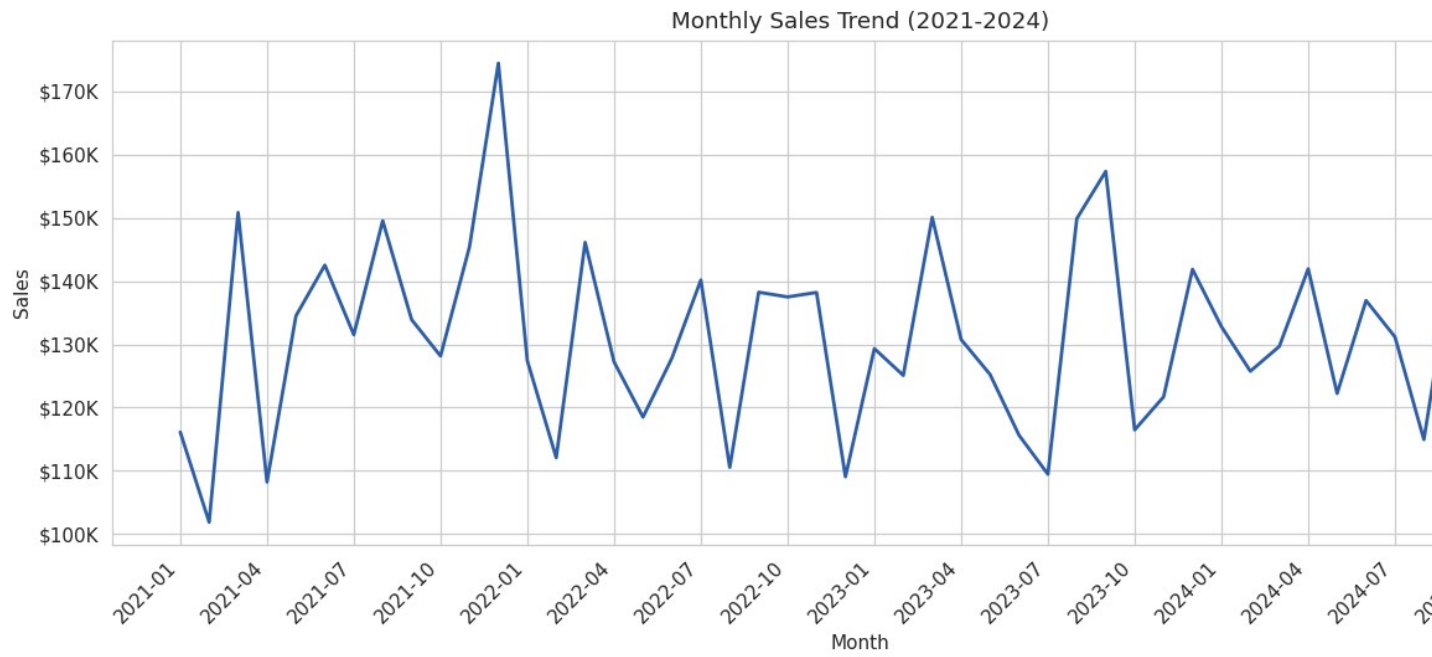


Figure 6. Monthly Sales Trend

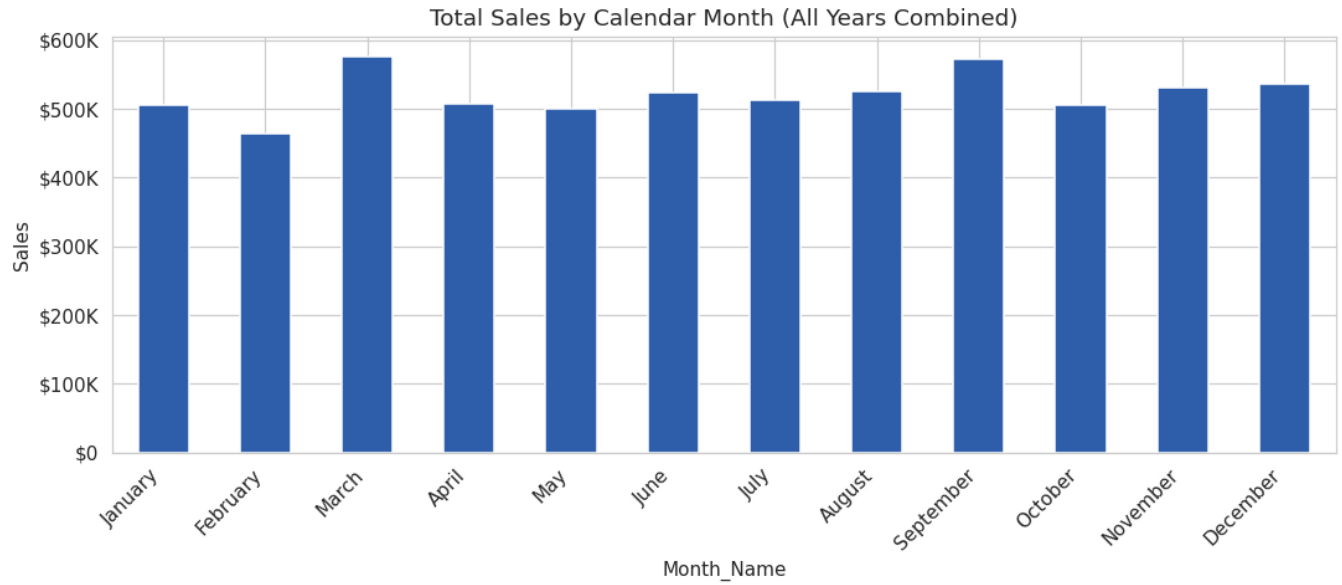


Figure 7. Sales by Calendar Month (All Years Combined)

6.7 Customer Analytics

Only 3 of the top 10 customers by sales also rank in the top 10 by profit. About 61.4% of customers (~487 of 793) generate 80% of sales — moderate concentration, broader than the classic "20% of customers" rule.

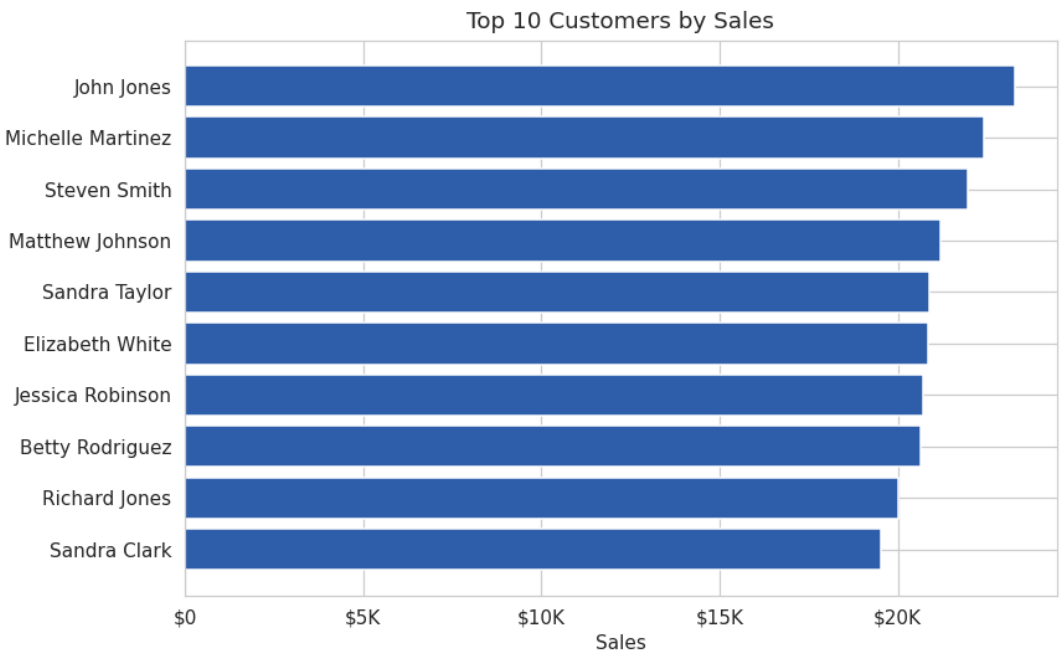


Figure 8. Top 10 Customers by Sales

6.8 Customer Segment Analysis

Segment	Sales	Profit	Customers	Profit Margin
Consumer	$3.17M	$268.0K	416	8.5%
Corporate	$1.78M	$164.3K	217	9.3%
Home Office	$1.32M	$101.1K	160	7.7%

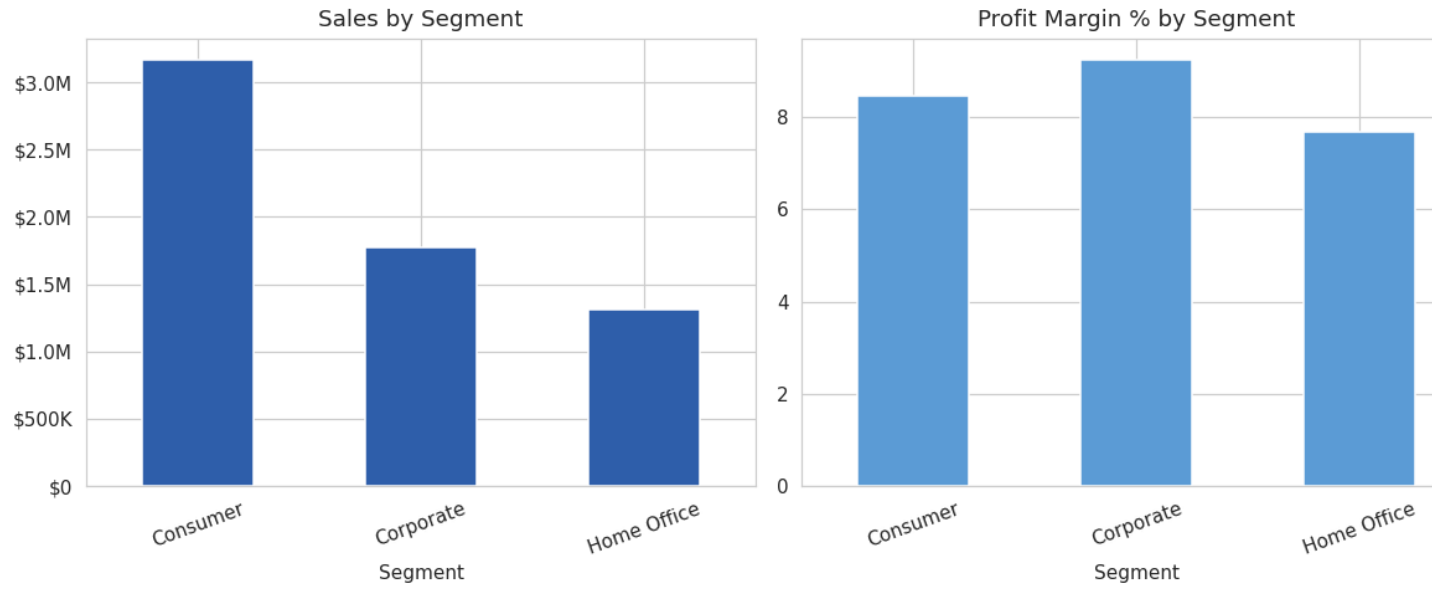


Figure 9. Segment Sales and Profit Margin

6.9 Shipping Analysis

Average delivery ranges from 0.5 days (Same Day) to 5.5 days (Standard Class). Profit margin is broadly similar across shipping modes — shipping speed is an operational lever, not a profitability lever, in this dataset.

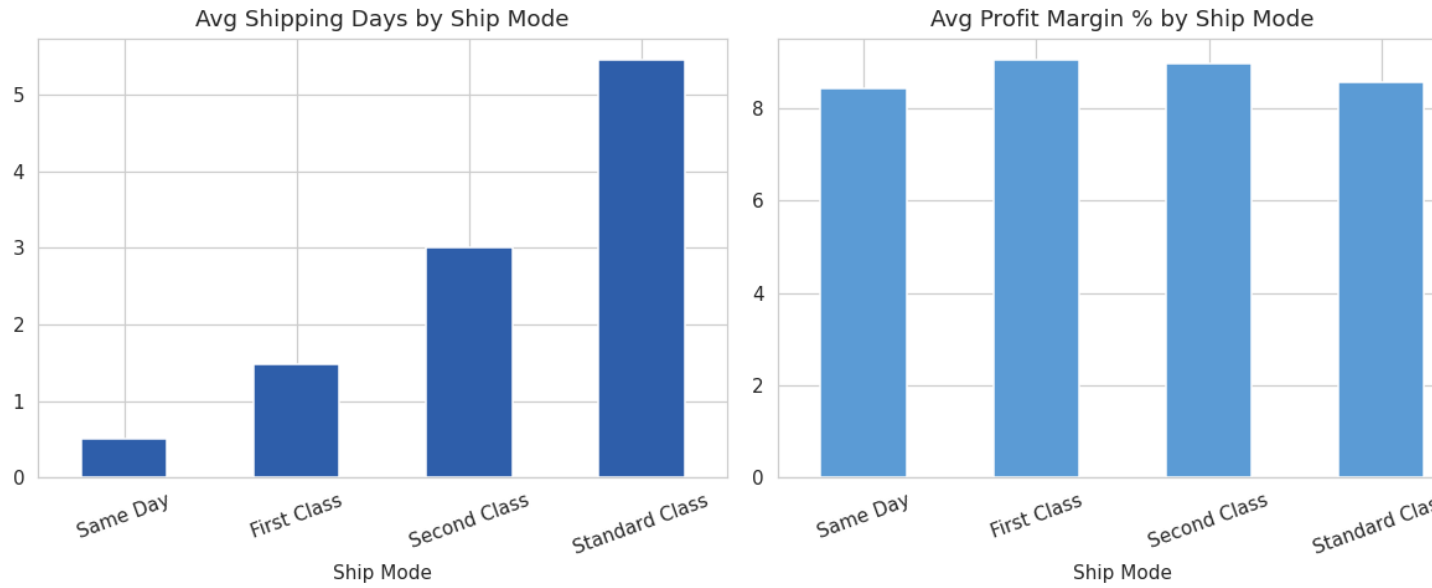


Figure 10. Shipping Time and Profit Margin by Mode

6.10 Executive Dashboard

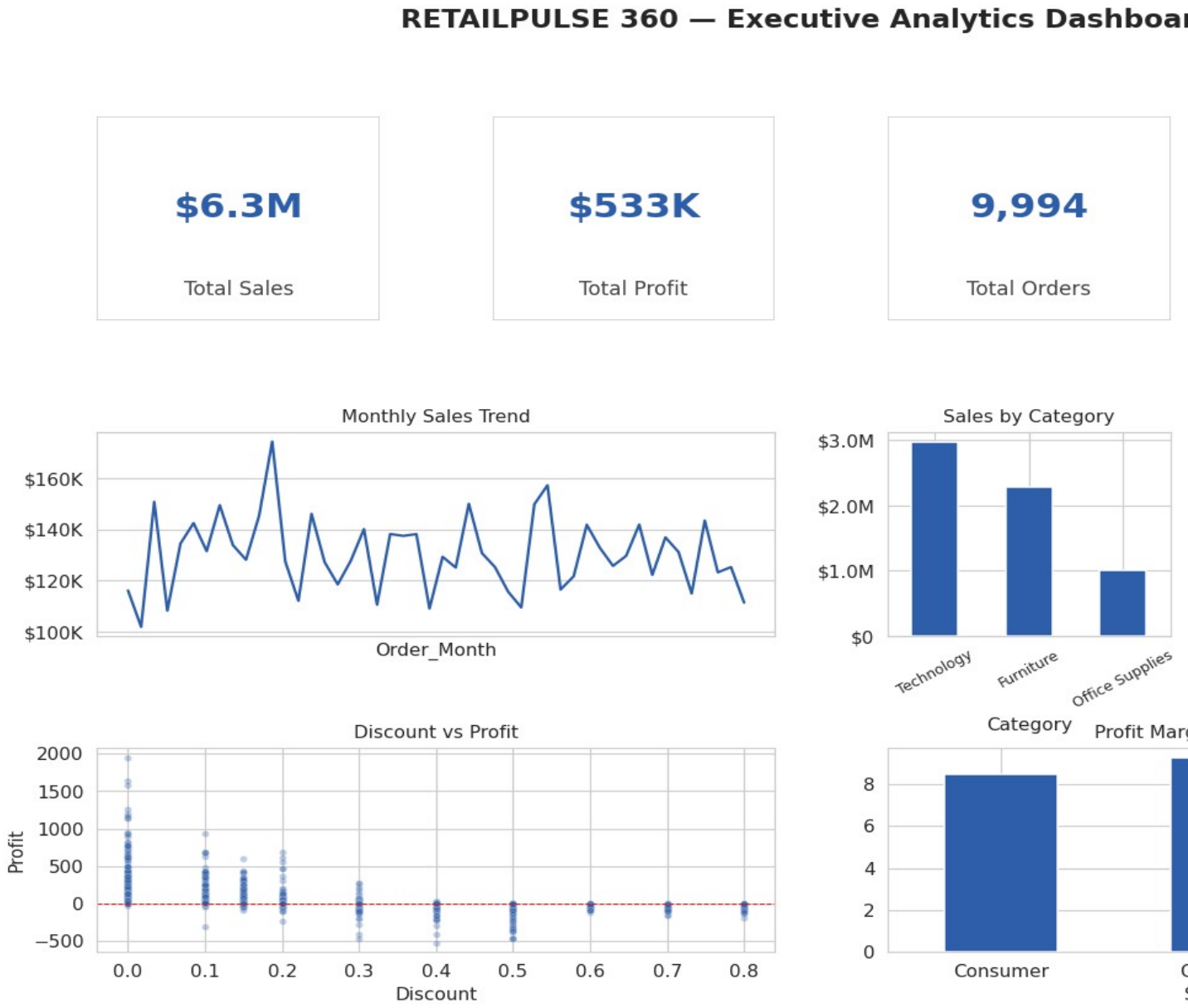


Figure 11. RetailPulse 360 Executive Analytics Dashboard

7. Key Findings

1. Technology drives both sales and profit leadership.
$2.97M in sales and $391.0K in profit — the highest of any category on both measures.

2. Furniture sells nearly as much as Technology but returns far less profit.
Furniture profit ($44.5K) is only a fraction of Technology's ($391.0K) despite comparable sales volume.

3. Deep discounts are reliably associated with unprofitable orders.
Correlation of -0.344 between discount and profit; average profit turns negative around 30% discount.

4. 29 products lose money overall, almost entirely Tables and Chairs.
The top 10 loss-makers are all Furniture items carrying above-average discounts.

5. Sales are flat year over year.
Only -4.8% change in yearly sales from 2021 to 2024.

6. There is a clear seasonal sales pattern.
March is the strongest month, February the weakest.

7. Regional performance is balanced.
No single region stands out as a problem or a standout winner.

8. Top sales customers are often not top profit customers.
Only 3 of the top 10 by sales also rank top 10 by profit.

9. Customer concentration is moderate.
61.4% of customers generate 80% of sales.

10. Shipping speed does not meaningfully affect profitability.
Profit margin is similar across all shipping modes.

8. Business Recommendations

Recommendation 1: Cap discount approval on Furniture at 20%.

Evidence: The 10 biggest loss-making products are all Furniture items carrying above-average discounts; average profit turns negative near 30% discount.

Recommended Action: Require manager approval for Furniture discounts above 20%, particularly on Tables and Chairs.

Recommendation 2: Run a pricing/cost review on the top loss-making Tables and Chairs SKUs.

Evidence: These products lose a combined -$7.7K, with the loss persisting across meaningful sales volume.

Recommended Action: Re-check landed cost and list price on these specific SKUs within 30 days.

Recommendation 3: Build a profit-based top-customer program.

Evidence: Only 3 of the top 10 sales customers are also top 10 by profit.

Recommended Action: Re-rank key accounts by profit contribution and route account management accordingly.

Recommendation 4: Prioritize Technology in marketing and inventory planning.

Evidence: Technology leads both sales ($2.97M) and profit ($391.0K) with the strongest margin of any category.

Recommended Action: Shift promotional budget and safety stock toward Technology ahead of the March peak.

Recommendation 5: Plan inventory and staffing around the observed seasonal peak.

Evidence: March is consistently the strongest sales month; February the weakest.

Recommended Action: Increase stock and staffing ahead of the peak month; reduce ahead of the trough.

9. Limitations

- This dataset represents historical transactions only and cannot predict future demand shifts.
- The discount-profit relationship is an association, not proven causation — discount level correlates with category and pricing decisions.
- External factors (competitor pricing, marketing spend, macroeconomic conditions) are not included.
- Customer satisfaction and service-quality data are unavailable, limiting shipping-mode conclusions to time and profit metrics only.
- The dataset covers a fixed historical period; seasonality and trend conclusions should be re-validated as new data arrives.

10. Conclusion

RetailPulse 360 converted raw retail transaction data into a structured view of sales, profitability, customers, products, regions, time trends, discounting and shipping. The clearest pattern is that sales volume and profitability move independently — the categories, regions and customers that sell the most are frequently not the ones that earn the most, and discounting past roughly 30% reliably erases profit. The five recommendations above give management a concrete, evidence-based path to improving profit without relying on sales growth, which the four-year trend shows cannot be assumed.